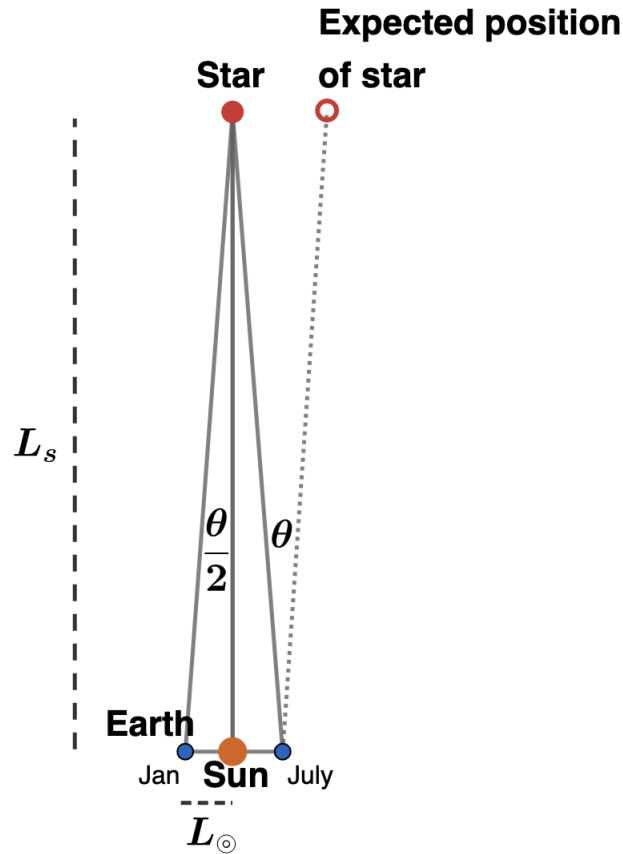


2023 F=ma Exam: Problem 23

Kevin S. Huang



Because of parallax, there is an angle difference in the apparent position of the star between January and July. We have

$$\tan \frac{\theta}{2} = \frac{L_\odot}{L_s}$$

Using the small angle approximation $\tan \theta \approx \theta$ and that $\theta = 2.5'' - 2'' = 0.5'' = (0.5)\left(\frac{1}{3600}\right)\left(\frac{\pi}{180}\right) = 2.4 \times 10^{-6}$, we find

$$\frac{L_s}{L_\odot} = \frac{2}{\theta} = \frac{2}{2.4 \times 10^{-6}} = 8 \times 10^5$$

so the answer is D.